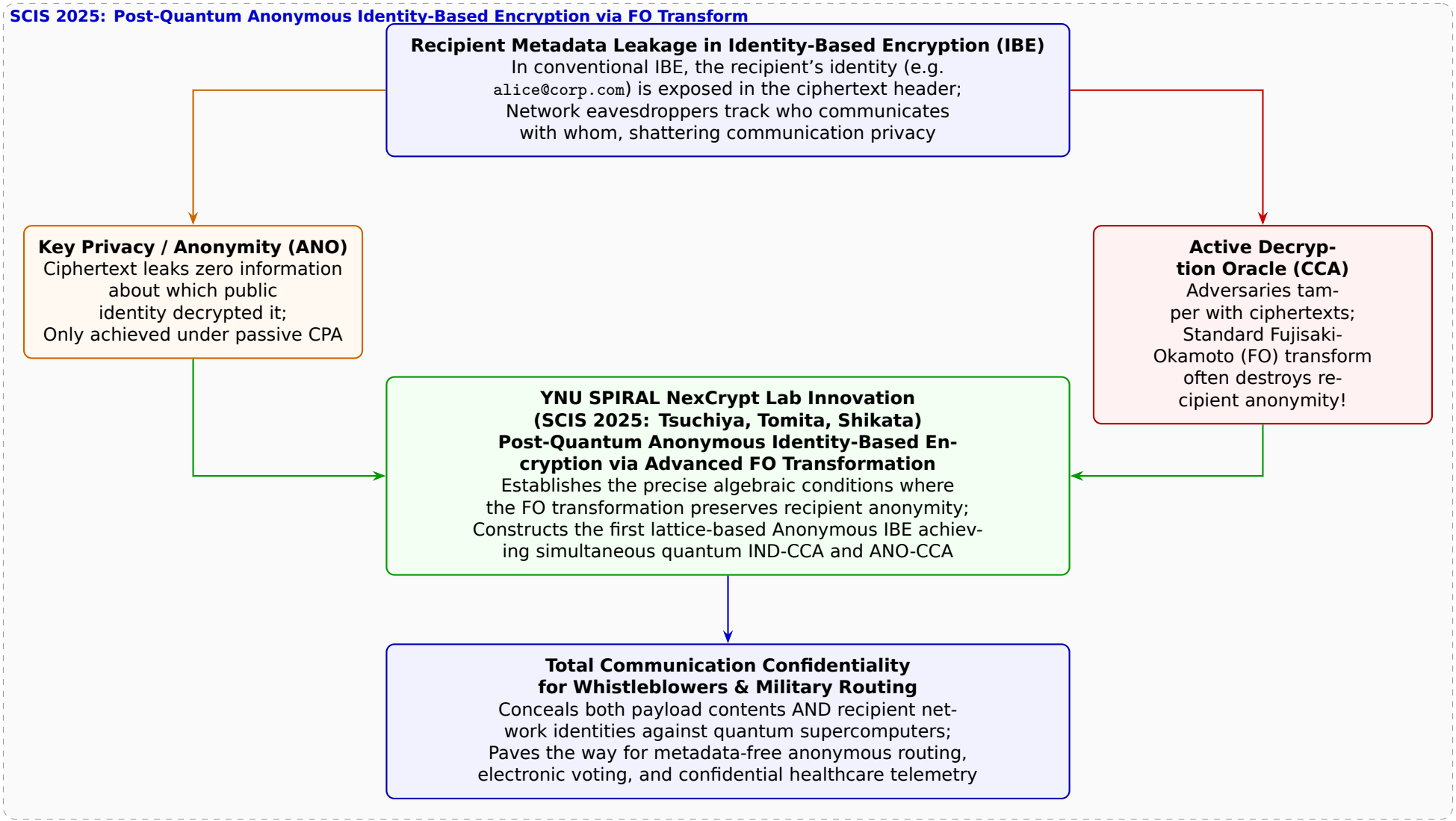
SCIS 2025: Post-Quantum Anonymous Identity-Based Encryption via FO Transform
Recipient Metadata Leakage in Identity-Based Encryption (IBE)
In conventional IBE, the recipient's identity (e.g. alice@corp.com) is exposed in the ciphertext header; Network eavesdroppers track who communicates with whom, shattering communication privacy
Key Privacy / Anonymity (ANO)
Ciphertext leaks zero information about which public identity decrypted it; Only achieved under passive CPA
Active Decryption Oracle (CCA)
Adversaries tamper with ciphertexts; Standard Fujisaki-Okamoto (FO) transform often destroys recipient anonymity!
YNU SPIRAL NexCrypt Lab Innovation
(SCIS 2025: Tsuchiya, Tomita, Shikata)
Post-Quantum Anonymous Identity-Based Encryption via Advanced FO Transformation
Establishes the precise algebraic conditions where the FO transformation preserves recipient anonymity; Constructs the first lattice-based Anonymous IBE achieving simultaneous quantum IND-CCA and ANO-CCA
Total Communication Confidentiality for Whistleblowers & Military Routing
Conceals both payload contents AND recipient network identities against quantum supercomputers; Paves the way for metadata-free anonymous routing, electronic voting, and confidential healthcare telemetry